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Abstract

Neuropsychiatry is rapidly evolving, seeking to bridge radiographic imaging and molecular pathology through machine-learning classification models. This study investigates the feasibility of leveraging class-specific autoencoders to extract latent spatial features associated with O6-methylguanine-DNA methyltransferase (MGMT) promoter methylation status in patients with glioblastoma multiforme (GBM). The framework consists of a 2D ResNet backbone that extracts spatial features from each MRI slice, followed by a Long Short-Term Memory (LSTM) layer that models inter-slice temporal dependencies. Enhancing MRI inputs with filter-derived activation maps yielded modest yet meaningful performance gains. Although preliminary, our findings illustrate how combining unsupervised representation learning with supervised classification can uncover imaging biomarkers. This approach offers a promising step toward more interpretable, data-efficient, and clinically relevant machine learning models in neuro-oncology.

Keywords (separated by '-')

CNN - Feature Extraction - MGMT Methylation - LSTM - Autoencoder - Glioblastoma

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Leveraging Autoencoder-Based Filters to Enhance Automated Medical Classifiers

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Abstract. Neuropsychiatry is rapidly evolving, seeking to bridge radiographic imaging and molecular pathology through machine-learning classification models. This study investigates the feasibility of leveraging class-specific autoencoders to extract latent spatial features associated with O6-methylguanine-DNA methyltransferase (MGMT) promoter methylation status in patients with glioblastoma multiforme (GBM). The framework consists of a 2D ResNet backbone that extracts spatial features from each MRI slice, followed by a Long Short-Term Memory (LSTM) layer that models inter-slice temporal dependencies. Enhancing MRI inputs with filter-derived activation maps yielded modest yet meaningful performance gains. Although preliminary, our findings illustrate how combining unsupervised representation learning with supervised classification can uncover imaging biomarkers. This approach offers a promising step toward more interpretable, data-efficient, and clinically relevant machine learning models in neuro-oncology.

AQ1 AQ2 AQ3

Keywords: CNN · Feature Extraction · MGMT Methylation · LSTM · Autoencoder · Glioblastoma

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1 Introduction

Timely and accurate diagnosis is essential in both clinical and operational environments, directly influencing patient outcomes and mission readiness. Developing reliable, non-invasive diagnostic tools remains a key goal across neuropsychiatric and neuro-oncological disorders. In this context, machine learning (ML) and deep learning (DL) have emerged as powerful methods for extracting clinically relevant patterns from high-dimensional neuroimaging data [16,17].

This study investigates the hypothesis that advanced machine learning models can detect latent, spatially complex signatures within magnetic resonance imaging (MRI) scans that correlate with molecular and pathological phenotypes.

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Such signatures, if validated, may serve as imaging biomarkers, enabling earlier, more precise diagnosis and longitudinal monitoring of disorders that otherwise lack clear phenotypic markers [1,6].

To evaluate this hypothesis, we focus on a well-characterized task in neuro-oncology: the classification of O6-methylguanine-DNA methyltransferase (MGMT) promoter methylation status in patients with glioblastoma multiforme (GBM). Glioblastoma is the most common and aggressive primary brain tumor in adults, with a median overall survival of approximately 12 to 15 months despite standard-of-care treatment [19,21]. MGMT promoter methylation plays a critical role in therapeutic response, as it suppresses DNA repair activity and increases tumor sensitivity to alkylating chemotherapeutics like temozolomide [8]. However, assessing methylation typically requires invasive tissue sampling and complex molecular assays. Recent studies have reported that MRI scans may capture indirect but informative features of MGMT methylation, potentially enabling non-invasive prediction through data-driven approaches [14].

By leveraging this well-studied molecular target, our research aims to bridge radiographic imaging and molecular pathology using deep learning-based classification models. Ultimately, this work serves as a stepping stone toward broader applications in neuropsychiatry, including the identification of imaging biomarkers for conditions such as Post-Traumatic Stress Disorder (PTSD) and Traumatic Brain Injury (TBI), where ground truth is similarly difficult to ascertain.

2 Related Works

Early efforts to noninvasively predict MGMT promoter methylation status from brain MRI data relied heavily on radiomics pipelines—leveraging manually segmented tumor regions, handcrafted feature extraction, and classical machine learning classifiers. These human-in-the-loop approaches identified statistical, textural, and morphometric features from MRI scans and applied models such as support vector machines (SVM), k-nearest neighbors (KNN), and ensemble methods to distinguish methylated from unmethylated glioblastomas.

Korfiatis et al. pioneered this space in 2016, extracting nine texture features from T2-weighted MRI and applying SVMs, achieving an AUC of 0.85 on a cohort of 155 patients [14]. Kanas et al. subsequently used a KNN model trained on twelve radiomic features from 86 patients and achieved 73.6% accuracy after dimensionality reduction [12]. In 2018, Xi et al. incorporated manual tumor segmentation by radiologists and extracted 30 features from T1 and T2 images, achieving an 86.6% classification accuracy using an SVM [23]. More recently, Do et al. utilized a combination of histogram, shape, and transform-based features with a random forest classifier, outperforming both XGBoost and SVM baselines and achieving 92.5% accuracy [5]. Other studies echoed these results: Le et al. applied XGBoost with an F-score-based feature selection to 53 patients, yielding an 88.7% accuracy [15], while Crisi et al. reported an AUC of 0.84 using SVMs [3].

These studies demonstrated that radiomic features—particularly those extracted from segmented tumor subregions—contain predictive signal regarding

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MGMT methylation. However, generalizability remains limited due to small sample sizes, lack of external validation, and dependence on expert-defined inputs. The need for consistent preprocessing and imaging protocols further restricts cross-site reproducibility.

To address these limitations, more recent research has transitioned toward automatic, deep learning-based methods. Convolutional neural networks (CNNs) have been employed to learn hierarchical features directly from raw MRI scans, either through whole-brain analysis or region-of-interest (ROI) segmentation. Pálsson et al. investigated a combination of radiomic and VAE-derived latent features, but found that radiomics alone achieved the highest AUC (0.632) with a random forest classifier [20].

ROI-based deep learning approaches have also shown mixed results. Capuozzo et al. applied CNNs to tumor-segmented regions but reported an AUC of just 0.58 [2]. Das et al. proposed an Intermediate State Generator (IS-Gen) to merge multiple MRI sequences into single-channel images for each patient, achieving an AUC of 0.66 using a ResNet-10 model [4]. Iranmehr et al. introduced temporal and spatial attention mechanisms to localize ROIs across FLAIR and T1wCE sequences before feeding them into an LSTM model, yielding an AUC of 0.70 [11].

Generalization remains a significant challenge for CNN-based methods, particularly when trained on institution-specific datasets. Kim et al. demonstrated that a SEResNeXt50 model trained on a small dataset from SNUH generalized poorly to BraTS, achieving an AUC of only 0.55 [13]. Faghani et al. showed that whole-brain input outperformed ROI-based strategies using a 3D DenseNet121, though even their best model reached only 0.65 AUC on T2-weighted scans [7]. Most recently, Hu et al. used a Vision Graph Neural Network (ViG) to analyze only the tumor's largest cross-sectional slice, obtaining an AUC of 0.63 [10].

Collectively, these studies highlight a persistent trade-off in the literature. Classical radiomics methods are interpretable and effective in small-cohort scenarios but often fail to scale across imaging centers. Deep learning models, though more expressive, require large and diverse datasets to avoid overfitting and capturing spurious correlations. The BraTS 2021 challenge exposed this gap by showing that most pre-2021 models exhibited dramatic drops in performance on standardized external data, with post-challenge AUCs averaging only 0.63×0.05.

These findings motivate a new generation of approaches that fuse biological insight with computational modeling. By integrating domain knowledge—such as features reflecting tumor cell density, necrosis, or bloodâĂŞbrain barrier breakdown—into either handcrafted pipelines or the inductive biases of deep models, researchers aim to produce representations that are not only accurate but also generalizable across diverse clinical settings.

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3 Methodology

To investigate the feasibility of leveraging learned representations for MGMT promoter methylation classification, we formulated a pipeline grounded in prior advances in deep representation learning and neuroimaging biomarker discovery.

3.1 Preprocessing and Dataset Construction

All multi-parametric MRI scans underwent standardized preprocessing. Each image was resampled to an isotropic resolution of 1 mm^3 to ensure consistent spatial granularity across modalities. FLAIR scans served as the anatomical reference, and all other modalities—T1-weighted (T1w), contrast-enhanced T1w (T1wCE), and T2-weighted (T2)—were rigidly co-registered to achieve voxelwise alignment. This preprocessing ensured that convolutional filters could operate on spatially aligned features, an essential assumption given that MGMT methylation is hypothesized to manifest as coherent radiographic signatures. Figure 1 shows the alignment across all four modalities for one slice of the same patient.

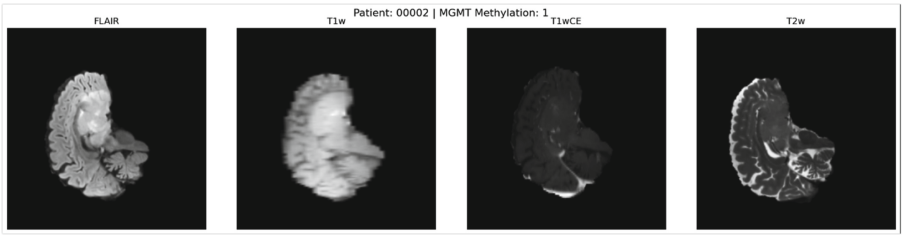

Fig. 1. Preprocessed Images

From the original BRaTS samples, two datasets were constructed:

- A methylated-only dataset, used to train an Autoencoder as an unsupervised feature extractor. (n = 202 patients, 35,956 images)
- A balanced classification dataset of methylated to unmethylated cases, used for downstream classification and evaluation. (n = 144 patients, 25,632 images)

3.2 Autoencoder Training on MGMT-Methylated Data

We trained a 2D convolutional autoencoder on individual MRI slices from the methylated-only dataset. The autoencoder architecture included a series of convolutional and pooling layers that compressed input data into a latent bottleneck representation, followed by symmetric decoder layers to reconstruct the original image. The goal was to learn compact encodings that capture structural features that could be uniquely associated with MGMT methylation [9, 22].

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By training exclusively on methylated cases, the autoencoder was encouraged to overfit to this class, ideally encoding methylation-relevant features into its latent space. As reconstruction loss minimized, we hypothesized that salient imaging biomarkers were being encoded in the internal filters and latent activations. Some sample filters are shown in Fig. 2.

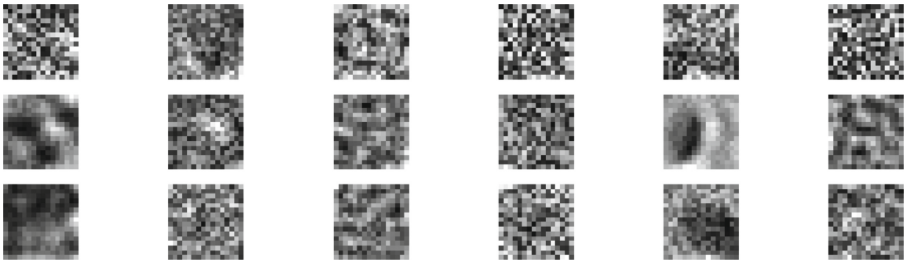

Fig. 2. Filters After Autoencoder Training

To visualize these internal encodings, we applied t-distributed stochastic neighbor embedding (t-SNE) to the latent vectors. This nonlinear dimensionality reduction technique projects high-dimensional data into two dimensions while preserving local similarity, making it suitable for identifying emergent clusters in medical imaging applications [18].

3.3 Filter Map Selection and Interpretation

To link the learned latent features to biologically relevant imaging patterns, we computed the center of mass of methylation-class clusters in the t-SNE space and backtraced them to specific convolutional filters. Among these, Filter Map 66, shown in Fig. 3, consistently showed high activation in regions corresponding to methylated tumors. Activation heatmaps indicated spatially coherent patterns potentially reflective of molecular or histopathological features.

3.4 Feature Fusion via Custom Combination Function

To incorporate the selected filter map into classification, we devised a custom fusion function. This procedure aimed to emphasize features associated with MGMT methylation while minimizing distortion of the anatomical context. For each voxel in the original MRI slice we compare its intensity with the intensity of the corresponding pixel in the selected filter map. If the difference is below a predefined similarity threshold, we replace the voxel value with the average of the two; otherwise we keep the original intensity unchanged.

By using the average value, this retains some of the original values while still incorporating the effects of the filter map. As seen in Fig. 4, there is a slight brightening effect in the image where the chosen filter is fused on top of the

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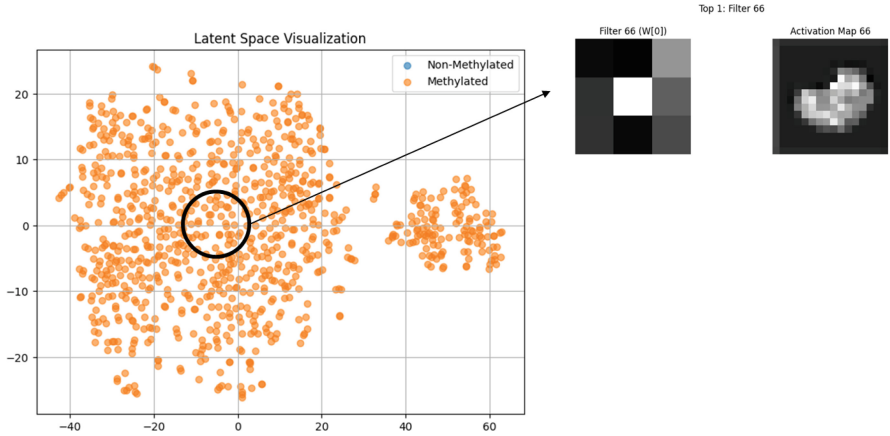

Fig. 3. Latent Space and Filter Map Selection

original image while much of the rest of the image is unchanged. Images of patients who have MGMT biomarkers experience a great brightening effect than those without.

3.5 Classification and Performance Evaluation

We used a custom 2D ResNet-LSTM hybrid architecture to classify MGMT promoter methylation status. The architecture starts with a 2D ResNet layer that extracts spatial features from each MRI slice, followed by a Long Short-Term Memory (LSTM) layer that models inter-slice temporal dependencies.

First, we established a baseline (Baseline) by training the model on the original, unmodified MRI slices. Then, we trained a new model after using our selected filter map to enhance the input images (Enhanced). This design allowed us to isolate the contribution of the latent feature to classification performance.

3.6 Evaluation Metrics and Experimental Design

We used the Area Under the Receiver Operating Characteristic Curve (AUC) as the primary metric of classification performance, as it reflects the trade-off between sensitivity and specificity.

Model robustness was evaluated via 5-fold cross-validation, ensuring consistent class balance across training and test folds. Early stopping based on validation loss was used to prevent overfitting. For each modality, we reported the average AUC across all folds and compared it to the baseline to assess performance impact.

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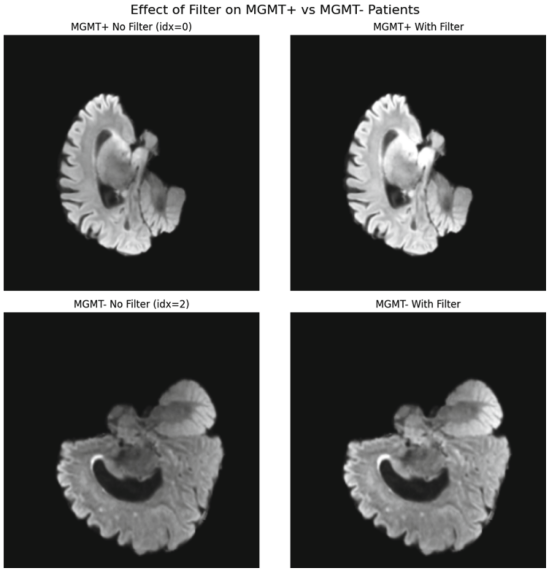

Fig. 4. Images Before And After Morphing

4 Quantitative Visualization and Performance Evaluation

To assess the discriminative impact of the enhanced dataset, we conducted a quantitative analysis of image-level intensity distributions. Each MRI slice was projected onto a two-dimensional plane defined by its mean voxel intensity (x-axis) and standard deviation of voxel intensities (y-axis). These metrics were chosen because we hypothesize that filter-map amplification will:

- Increase the mean intensity, reflecting brighter regions due to feature amplification.
- Increase the standard deviation, reflecting greater heterogeneity in voxel values consistent with class-specific spatial variation.

Figure 5 visualizes this comparison. In the original dataset, slices from both classes cluster tightly in the lower-left quadrant, indicating limited intra-class variance and overlap between methylated and unmethylated samples. This compression suggests poor separability using standard statistical thresholds. Conversely, the enhanced dataset exhibits increased spread and emerging stratification between classes—indicating that the inclusion of filter-derived activations introduces class-relevant intensity variability. Although the classes do not form fully separable clusters, this increased dispersion supports the hypothesis that the filter map adds discriminative signal.

To further examine modality-specific sensitivity, we repeated the experiments independently across the four common MRI modalities: T1w, T1wCE, FLAIR, and T2w. These results are shown in Table 1. Key observations include:

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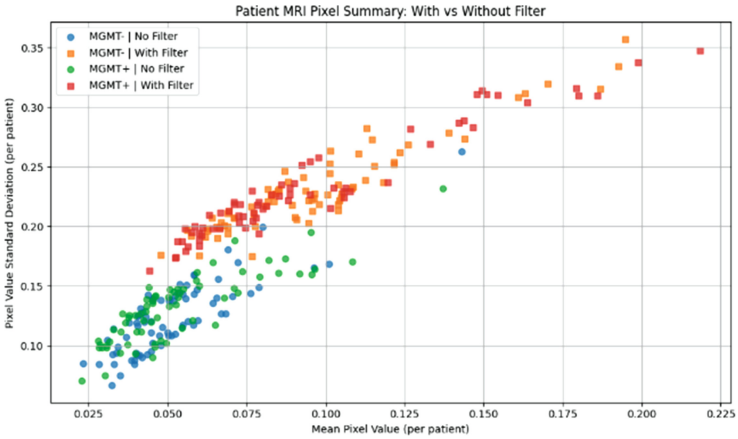

Fig. 5. Original versus Enhanced Pixel Summary

- T2-weighted images showed the largest performance gain when enhanced with the filter map, suggesting a stronger sensitivity to methylation-associated structural variation in this modality.
- FLAIR images demonstrated negligible performance improvement post-enhancement, with AUC differences below statistical significance, indicating a potential mismatch between the enhanced features and modality-specific contrast.

Table 1. AUC scores for baseline and filter-enhanced models.

	FLAIR	T1w	T1wCE	T2w	*All*
AUC - Baseline	0.64	0.64	0.61	0.61	*0.63*
AUC - Enhanced	0.65	0.67	0.65	0.69	*0.66*

5 Discussion

Our findings demonstrate that training an autoencoder exclusively on MGMT-methylated glioblastoma cases can uncover spatial features that, when reintroduced via filter-based enhancement, lead to modest but measurable improvements in classification performance of a 2D ResNet/LSTM hybrid architecture. The most notable gains were observed in T2-weighted MRI scans, suggesting that class-specific representation learning can encode clinically relevant structural variations associated with MGMT promoter methylation.

However, these improvements were not consistent across imaging modalities. For instance, the FLAIR modality exhibited minimal enhancement, indicating

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either that the selected filter lacked generalizability or that FLAIR images are inherently less sensitive to MGMT methylation-related features in this context. This variation highlights the importance of modality-aware modeling strategies in multi-parametric neuroimaging pipelines.

When compared to prior literature, our enhanced T2-weighted model (AUC = 0.69) outperformed several state-of-the-art approaches, including the 3D DenseNet121 reported by [7] (AUC = 0.65). Additionally, there was a slight improvement over the post-BraTS 2021 challenge average (AUC = 0.63 ± 0.05), these results demonstrate that our approach achieves competitive performance relative to established benchmarks and previously reported models.

Our hybrid ResNet-LSTM classifier, designed to capture both spatial and contextual (slice-wise) dependencies, effectively leveraged the enhanced representations. Although the filter-derived augmentation did not dramatically increase class separability, it introduced additional signal that complemented the base model. In clinical contexts such as neuro-oncology—where early, accurate molecular characterization is critical—even incremental improvements can translate to meaningful benefits in diagnosis and treatment planning.

Crucially, this study provides a proof of concept on how autoencoders can be repurposed beyond mere input reconstruction: they can serve as targeted feature generators, identifying latent representations aligned with specific clinical attributes. This modular, interpretable approach bridges unsupervised and supervised learning, supporting more transparent and biologically motivated models in medical AI.

5.1 Limitations and Future Work

Several limitations of this study merit consideration. First, the relatively limited dataset size constrains the generalizability of our findings. While cross-validation was used to mitigate overfitting, the robustness of the proposed pipeline must be validated on larger, more heterogeneous datasets—ideally spanning multiple institutions and acquisition protocols—to assess its real-world applicability.

Second, the selection of a representative filter map was performed manually, introducing subjectivity. Although guided by empirical activation patterns, this process introduces subjectivity and may overlook latent features with equal or greater relevance. Future studies should adopt automated filter attribution techniques, such as gradient-based saliency methods or attention-weighted visualization, to support more systematic and reproducible feature discovery.

Looking ahead, integrating auxiliary clinical metadata—such as age, tumor location, or prior treatment history—could enrich the model's contextual understanding and improve both performance and interpretability. The fusion of imaging-derived biomarkers with structured clinical data represents a promising direction for developing comprehensive, non-invasive diagnostic tools that align more closely with clinical decision-making processes. Additionally, we would like to apply similar work for analysis on other problematic areas such as PTSD, TBI, or CTE.

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5.2 Conclusion

This study presents a novel framework for leveraging class-specific autoencoders to extract latent spatial features associated with MGMT promoter methylation in glioblastoma. By enhancing MRI inputs with a filter maps, we observed modest but meaningful improvements in classification performance—most notably in T2-weighted images. While preliminary, our findings demonstrate the potential of combining unsupervised representation learning with supervised classification to uncover imaging biomarkers. This approach offers a promising step toward more interpretable, data-efficient, and clinically relevant machine learning models in neuro-oncology.

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Chapter 27

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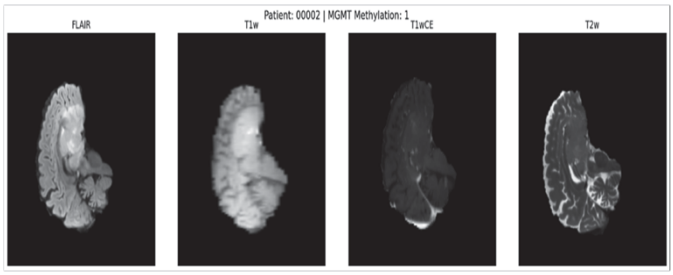
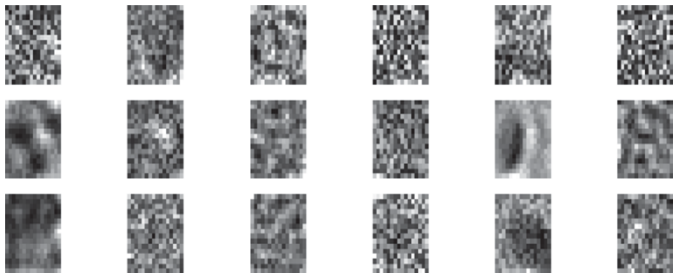
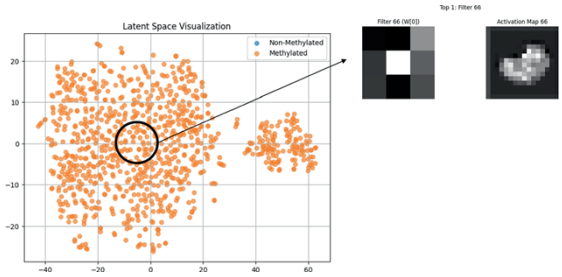
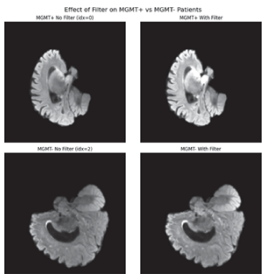
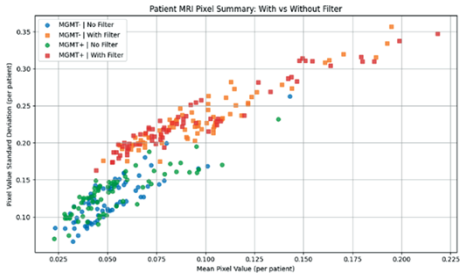
Page no	Fig/Photo	Thumbnail	Alt-text Description
4	Fig1		Four-panel fluorescence microscopy image showing different views of a biological sample. Each panel displays varying levels of brightness and contrast, highlighting distinct structural features. The panels are labeled with different parameters, indicating variations in imaging conditions or sample treatments. The background is uniformly dark, enhancing the visibility of the sample details.
5	Fig2		Array of electron microscopy images showing various grayscale patterns and textures. Each image displays a distinct arrangement of light and dark areas, suggesting different structural or material compositions. The images are organized in a grid with two rows and eight columns.
6	Fig3		Scatter plot titled "Latent Space Visualization" showing two classes: non-methylated (blue) and methylated (orange) data points. A cluster of orange points is highlighted with a circle and arrow pointing to two images on the right. The first image is labeled "Filter 66 (W[0])" showing a black and white grid pattern. The second image is labeled "Activation Map 66" displaying a grayscale pixelated pattern.
7	Fig4		The image is a panel figure showing MRI scans of brain sections, illustrating the effect of a filter on MGMT positive and negative patients. The top row displays MGMT positive scans, with the left image

Page no	Fig/Photo	Thumbnail	Alt-text Description
			showing no filter and the right image with a filter applied. The bottom row shows MGMT negative scans, with the left image without a filter and the right image with a filter. The scans highlight differences in brain structure visibility with and without the filter.
8	Fig5		Scatter plot showing the relationship between model output and actual data. The chart includes four data series, each represented by different colored markers: blue, green, red, and orange. The x-axis is labeled "Model Output (calibrated)" and the y-axis is labeled "Fractional Bias (calibrated)." A legend identifies the data series as different models with and without filters. The data points generally trend upward, indicating a positive correlation.